



Original Article

Strengths and limitations of before–after–control–impact analysis for testing the effects of marine protected areas on managed populations

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Kerr, L. A., Kritzer, J. P., and Cadrin, S. X. Strengths and limitations of before–after–control–impact analysis for testing the effects of marine protected areas on managed populations. – ICES Journal of Marine Science, 76: 1039–1051.

Received 29 June 2018; revised 17 December 2018; accepted 15 January 2019; advance access publication 19 February 2019.

Marine protected areas (MPAs) are a common management strategy for conserving marine resources, but it can be challenging to evaluate their effectiveness for meeting management objectives. Measuring the effectiveness of MPAs is particularly challenging in dynamic and changing environments where other management approaches are simultaneously implemented. Before–after–control–impact (BACI) analysis is a tool that offers a simple and robust design for evaluating complex effects. However, design and interpretation of a BACI analysis is not always straightforward. The goal of this study was to explore the potential for BACI to evaluate MPA performance in a system simultaneously impacted by other management measures and environmental change. We develop a typology of interpretations of BACI results based on the main and interaction effects of the model, categorized by the extent to which dynamics inside and outside of the MPA are independent. Furthermore, we examine how decisions about the spatial and temporal design of the study, and the focal species and response variables, can determine which outcomes from within the typology are evident through BACI applications to New England groundfish area closures. We identify strengths and limitations of the BACI approach and demonstrate that BACI is a valuable but imperfect tool for evaluating MPAs.

Keywords: BACI, closed areas, haddock, marine protected areas, yellowtail flounder.

Introduction

Marine protected areas (MPAs) have been widely applied with the goal of achieving a variety of management objectives in marine ecosystems. These objectives can be related to a range of ecosystem services, including resource conservation, fisheries production, tourism, climate resilience, and others (Agardy *et al.*, 2003; Lubchenco *et al.*, 2003; Jennings, 2008; Liu *et al.*, 2018). Global reviews of MPA effects have documented significant increases in key biological variables, including biomass, density, organism size, and species richness. These studies are primarily based on before–after or inside–outside analyses (Halpern and Warner, 2003; Lester *et al.*, 2009; Edgar *et al.*, 2014). However, studies in which MPAs are monitored both before and after implementation and inside

and outside reserve boundaries are relatively rare (Lester *et al.*, 2009; Liu *et al.*, 2018). Such studies are valuable because they effectively control for natural ecosystem dynamics and variability on a regional scale and can help distinguish spurious effects resulting from spatial differences between sites (Lester *et al.*, 2009).

Before–after–control–impact (BACI) analysis (Green, 1979; Stewart-Oaten *et al.*, 1986) and its extensions (e.g. BACI Paired Series, beyond BACI, and Multiple BACI; Underwood, 1991, 1993) are considered to be the most rigorous designs for assessing the impact of MPAs (Ojeda-Martínez *et al.*, 2011; Sciberras *et al.*, 2013). The general BACI framework involves measurement of an ecological variable of interest at a site (e.g. density of a species) that is expected to be positively or negatively affected by a

management action or other anthropogenic activity, before and after the intervention, in comparison to one or more control sites that are not expected to be affected in the same manner (Green, 1979). BACI is particularly useful for assessing large-scale, unreplicated events, such as anthropogenic disturbances or large-scale environmental management interventions (Schwarz, 2013). These events can be viewed as experiments with uncertain outcomes that are challenging to replicate due to confounding effects of other management actions and/or environmental variability (Walters and Holling, 1990).

Measuring the effectiveness of MPAs is particularly challenging where other management approaches are simultaneously implemented, potentially confounding MPA effects from non-MPA effects. Similarly, dynamic environmental conditions that operate across large spatial scales (e.g. climate change) can also confound interpretation. BACI analysis can be a useful tool for MPA evaluation, but traditional applications and interpretations of results may not sufficiently account for ecological scales and processes relevant to managed populations. One limitation of a simple BACI sampling design is that the temporal dimensions are binary (i.e. before vs. after) and potentially important patterns within either time period cannot be readily distinguished (Underwood, 1992; Stewart-Oaten and Bence, 2001). Consideration of temporal processes and variance in BACI has been recommended, but is contingent on data availability (Stewart-Oaten *et al.*, 1986; Stewart-Oaten and Bence, 2001). Furthermore, although the framework can accommodate greater spatial complexity (i.e. by including multiple control and/or impact sites; Underwood, 1992), it is not well suited to addressing spatial interactions among sites, or environmental change experienced differently in different locations within or among sites. As a result, both false positives and undetected effects are possible when using this approach.

False positives and undetected effects can be avoided by analytical design and understanding the spatial and temporal scales of other management actions, environmental variation, and resulting ecological processes. For example, an apparent significant MPA effect in the BACI framework (i.e. a statistically significant model interaction) may reflect different stochastic environmental effects among control and impact sites. Including data from multiple years in the before and after periods can help to avoid temporarily confounding effects (Underwood, 1994). However, if environmental variation is directional (e.g. warming associated with climate change) rather than stochastic and experienced differently among sites in the analysis, interpretation of BACI outcomes will be complicated, even with multiple sampling years before and after implementation. For example, a nearshore MPA in the North Sea (termed the “Plaice Box”) designed to protect juveniles of a commercially important flatfish had fewer fish after it was created than before. This unexpected outcome has been attributed to increasing water temperature and an offshore shift in distribution of flatfish (van Keeken *et al.*, 2007). Because the Plaice Box encompassed almost the entire nearshore nursery grounds for plaice and those areas were affected differently by warming waters than offshore areas (van Keeken *et al.*, 2007), BACI would not have been an appropriate analytical framework as there were not suitable control areas that experienced similar environmental change. Furthermore, the spatial scale over which environmental variation is experienced similarly can change through time (Kelly *et al.*, 2009), so an appropriate analytical design at one time might not be appropriate at another time.

One major limitation of BACI for evaluation of MPAs, and any natural experiment of this type, is that there are no true controls comparable to those possible in a laboratory setting. Ideally, control areas are unaffected by the MPA, but are influenced by all other temporal factors exogenous to the MPA (e.g. environmental conditions; Stewart-Oaten and Bence, 2001). Another approach for reducing the likelihood of false positive or undetected effects is to include multiple control and impact sites to minimize the effect of stochastic variability among sites (Underwood, 1992). For example, Tissot *et al.* (2004) evaluated the performance of MPAs off Hawaii for recovery of aquarium fishes. Their study design included five locations with both control and impact sites, and they found patterns that were consistent among sites. Employing multiple sites expands the spatial domain of the study which can be powerful in allowing for generalization of MPA impacts across sites despite any ecological dissimilarity. The study by Tissot *et al.* (2004) included well-mixed control and impact sites, but other studies with fewer impact sites might experience greater constraints on the valid spatial scope and potential for replication.

Regardless of the number of control and impact sites and their spatial distribution, BACI assumes that the sites are independent (Underwood, 1991). Impact effects can be obscured by ecological interactions between the control and impact sites that homogenize the response (Claudet, 2018). Given that spillover of adult fish or export of planktonic larvae from MPAs to fishing grounds is a potential MPA outcome and design consideration (Halpern and Warner, 2003), the independence assumption of BACI needs careful consideration when evaluating MPA performance. In a study of an MPA in eastern Brazil, Francini-Filho and Moura (2008) addressed this issue by defining spillover as part of the impact being analysed, and therefore defined two sites outside of the MPA but within an expected spillover zone as impact sites, in addition to the MPA itself. Their study included only a single control site that was assumed to be far enough away from the MPA as to not experience spillover. However, this creative solution to the confounding effects of spillover implies additional assumptions, and requires a third treatment in the analysis to distinguish the protected area from the spillover zone. The potential for ecological interactions between MPA and non-MPA sites via spillover, larval dispersal, or other processes raises important questions about appropriate BACI design, interpretation of BACI outcomes, and even whether BACI is an appropriate analytical framework.

Rigorous evaluation of the effectiveness of MPAs for meeting their stated goals and objectives is needed, but global reviews indicate that assessments of effectiveness are limited (Pomeroy *et al.*, 2005; Ojeda-Martínez *et al.*, 2011). Although there may be unintended benefits of MPAs, it is important to recognize the original objectives when evaluating performance because there are tradeoffs in the design of MPAs to achieve different objectives (Halpern and Warner, 2003). MPA design and evaluation of MPA performance can be hindered by data availability as well as our understanding of ecological processes (Sale *et al.*, 2005). MPA implementation is rarely treated as a true adaptive management experiment, with well-designed sampling protocols tailored to the outcomes of interest and covering an adequate number of years before and after the change (Walters and Holling, 1990). Data collection programmes, such as fishery-dependent or fishery-independent monitoring of catch, abundance, population structure, or species diversity, can be used for BACI analysis (Jaworski *et al.*, 2006), but such opportunistic samples can also introduce analytical limitations or narrow the questions that can

be addressed. Dedicated sampling to evaluate MPAs is rarely initiated well before the MPA is established (Fraschetti *et al.*, 2002; Liu *et al.*, 2017). Appropriate analytical frameworks that can utilize available data and lend insight into performance are needed for successful implementation of MPAs.

The goal of this study was to explore a range of technical and ecological issues related to the potential for BACI to evaluate MPA performance in a system simultaneously impacted by other management measures and environmental change. We developed a typology of BACI outcomes and proposed likely ecological drivers underpinning each interpretation of results. We illustrate the application of the BACI outcome typology using data from fishery resource surveys in, and adjacent to, two MPAs in the waters off the Northeastern United States. In doing so, we examined the implications of alternative definitions of control sites based on different assumptions about spatial independence, as well as choice of response variable (different species and different population attributes) for determination of MPA effectiveness. We also examined effects of greater spatial and temporal complexity in ecological dynamics on the outcomes and interpretations of BACI analysis. Our results offer insights into the strengths and limitations of using BACI to evaluate performance of MPAs.

Material and methods

Development of typology of BACI outcomes

We developed a typology of BACI results based on temporal trends and the strength and sign of main effects and the time–

area interaction effect. Definition and proposed interpretation of these outcomes and the potential underlying processes driving each are derived from the theoretical and empirical literature on MPAs, fisheries, and marine ecology. We use examples from the case study analysis described below to illustrate these different BACI outcomes types.

Study system

A network of year-round MPAs was created in the US waters of the northwest Atlantic Ocean beginning in 1994, during a period when several principal groundfish species were considered collapsed and in need of rebuilding (NEFMC, 1994). The MPAs include two large areas on Georges Bank (Closed Areas I and II, Figure 1). Commercial fishing with bottom-tending gears is prohibited within these areas, with the exception of lobster traps and the limited access scallop dredge fishery, which is managed in the MPAs on a rotational basis (O’Keefe and DeCelles, 2013). Recreational hook-and-line fishing and midwater trawling are also allowed within these MPAs. Closed Areas I and II were originally designed to protect haddock spawning (NEFMC, 1994), but they are also considered to reduce fishing mortality on other groundfish species in the region (Sherwood and Grabowski, 2016). Over the period that these MPAs have been in effect, there have also been extensive changes in fisheries management within the region, as well as large-scale ecosystem change (Pershing *et al.*, 2015; Saba *et al.*, 2016). Previous studies have examined the

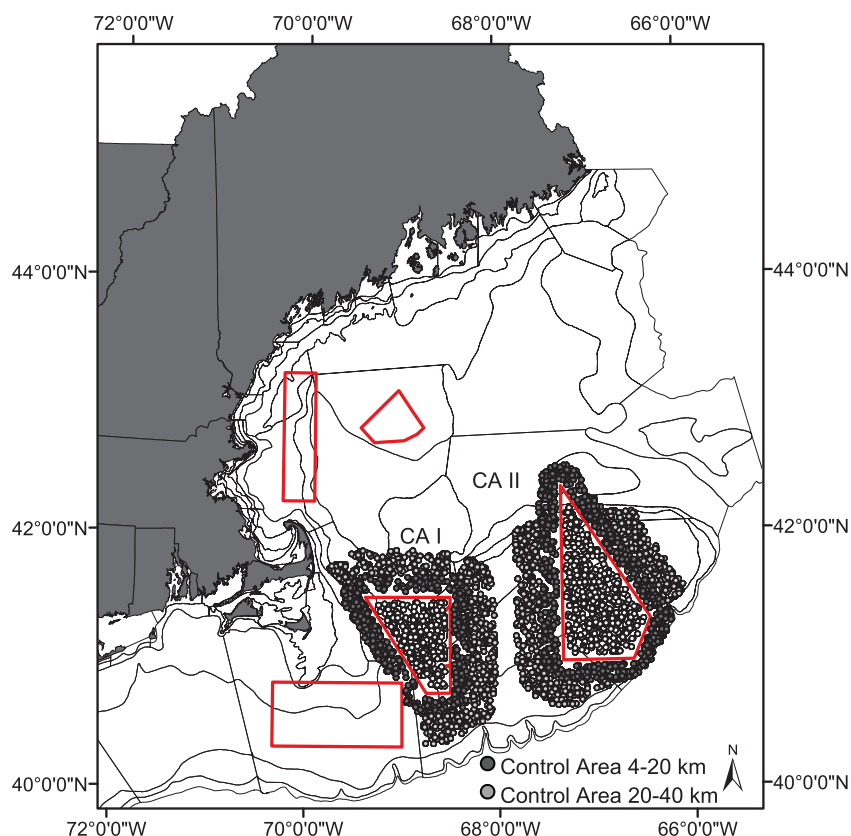


Figure 1. Map of closed areas in New England waters (outlined) with a focus on Georges Bank closures: Closed Area I (CAI) and Closed Area II (CAII). The map depicts that location of bottom trawl survey tows inside Closed Areas I and II (white circles) and in associated control areas defined at two alternative distances (4–20 km: dark gray circles and 20–40 km: light gray circles). Note, that data from the closed area boundary to a distance of 4 km are not used in this analysis to ensure independence of survey tows.

outcomes of these MPAs with respect to habitat (Collie *et al.*, 1997; Hermesen *et al.*, 2003; Lindholm *et al.*, 2004), sea scallop abundance (Stokesbury, 2002; Hart and Rago, 2006; Stokesbury and Harris, 2006), and groundfish abundance (Murawski *et al.*, 2005; Pereira *et al.*, 2012; Sherwood and Grabowski, 2016).

Data

Our analysis focused on evaluating the impact of Closed Areas I and II on two principal groundfish species managed under the New England multispecies plan (large-mesh groundfish): haddock (*Melanogrammus aeglefinus*) and yellowtail flounder (*Limanda ferruginea*). Haddock and yellowtail flounder were selected to test the effectiveness of closed areas relative to their original intent of haddock spawning protection and their ancillary benefit of reducing fishing mortality on another groundfish species in the region for which these areas are the principal habitat. Furthermore, haddock and yellowtail flounder present contrasts in habitat use (off-bottom vs. benthic) and their scale of movement (wider ranging vs. more sedentary) that might lead to different MPA responses.

We used data from the stratified-random bottom trawl survey conducted semiannually (spring and autumn) by the Northeast Fisheries Science Center (NEFSC; Azarovitz, 1981). Survey data from 1979 to 2010 were used in this analysis to include 16 years before and after implementation of the MPAs. Tow information consisted of timing and location of individual bottom trawl tows and species presence and absence information, as well as details on the specific gear configurations and vessel used at the time of data collection. The catch of each species was quantified in number and weight (kg) for each tow and these values were adjusted for differences in gear configuration and survey vessel over time (Miller *et al.*, 2010).

Data classification

We classified data from inside MPAs and control areas outside MPAs into time periods of equal length before (1979–1994) and after (1995–2010) establishment of MPAs in 1994. Geographic information system technology (ArcGIS, Version 10.3) was used to classify survey data spatially, identifying data collected inside each MPA and defining control areas for comparison. Data inside each of the MPAs were defined as survey tows that occurred within the MPA boundary, excluding tows that occurred within 4 km of the boundary. This 4 km distance was defined as a buffer directly both inside and outside of each MPA boundary from which no data were used. The 4 km distance encompasses the length of trawl survey tows, so that there is no potential overlap in tows used to characterize areas inside and outside of MPAs. The 4 km distance also equates to a region of spillover identified by Murawski *et al.* (2005), although we also considered the possibility that spillover operates over a larger scale.

We defined control areas that were outside of MPAs by restricting data selection in two ways. First, the selection was constrained to include only data from survey strata represented within MPAs. This constraint was designed to maximize comparability among control and impact sites to the degree possible, as survey strata are designed to encompass the same habitat type, as defined by depth and region. Survey strata that represented <2% of the total number of survey tows within the MPAs were not selected for control areas. Control areas were

further constrained to a geographic distance from the respective boundary of the MPA. We examined two alternative distances: 4–20 and 20–40 km. The former assumes spillover is limited to the 4 km halo estimated by Murawski *et al.* (2005), whereas the latter allows for more extensive movements (Cadrin, 2010). Murawski *et al.* (2005) analysed fishery data and found that most haddock catch was within 4 km of a closed area boundary and most yellowtail catch was within 5 km of a closed area boundary. Cadrin (2010) reviewed tag recovery information to infer yellowtail flounder movement patterns, and the most relevant study found that 85% of recaptures were within 20 km of the release location (Melgey, 2010). Data selection was not balanced by design due to data limitations, but the number of selected observations was approximately balanced among the four treatments (inside: before, inside: after, outside: before, outside: after) to provide power of detection and robustness to unequal variances.

Statistical approach

The BACI model included the main effects of location (inside the MPA and the control area) and time (before and after MPA implementation), and the interaction of location and time period. We fit models to abundance and biomass (kg) per tow data from the NOAA stratified-random bottom trawl survey using a zero adjusted inverse Gaussian distribution. These models incorporated a mixed discrete-continuous distribution that specified a logit-linear model for the occurrence of a fish (i.e. probability of fish presence) and a log-linear distribution for either mean abundance or biomass (Rigby and Stasinopoulos, 2007). This approach enabled us to deal with the large number of zeros and positive skewness that characterized the trawl survey data. The model was fit using the `gamlss()` function in the `gamlss` package with family = ZAIG (Rigby and Stasinopoulos, 2007) in R (R Development Core Team, 2016).

In assessing the performance of MPAs in New England, we compared results based on the choice of species (haddock and yellowtail flounder), response variable (probability of species occurrence, abundance per tow, and biomass per tow), and alternative definitions of control areas (defined by distance from the MPA boundary: 4–20 and 20–40 km). Because determining appropriate significance levels for large-scale environmental impact assessments like BACI analysis of MPA effects requires careful consideration of objectives, risk tolerance, and both Type I and Type II error rates (Mapstone, 1995), we discuss these results in terms of relative directional trends as well as significance determinations at the 0.05 level.

In addition, we examined other aspects of temporal and spatial complexity in MPA impacts. We calculated annual summary statistics [mean number per tow and weight per tow (kg)] inside and outside of MPAs and summarized data for individual survey strata for the 16-year periods before and after implementations of closures. We also used ArcGIS (ESRI) for spatial visualization of fish distribution and abundance at finer spatial scales before and after MPA implementation.

Results

Typology of BACI outcomes

Our typology includes traditional interpretations of BACI results and alternative interpretations that consider the complex responses of mobile organisms in dynamic and interconnected

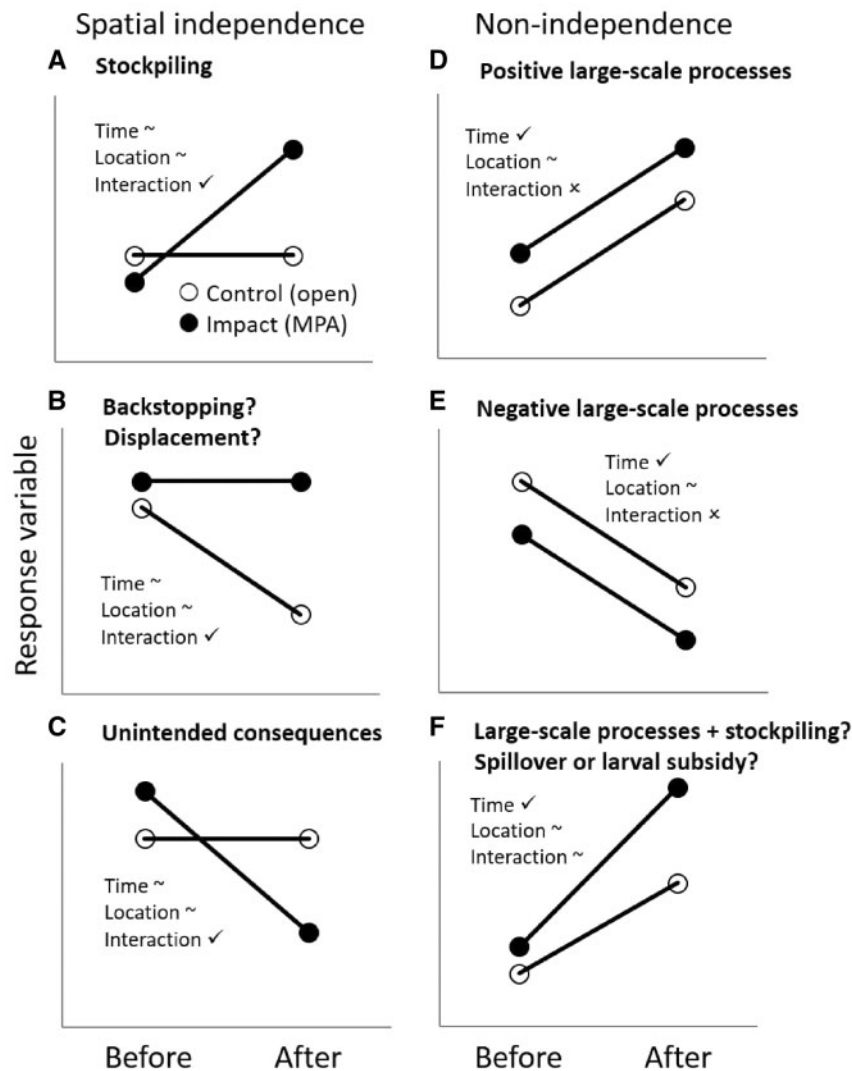


Figure 2. Generalized figure illustrating typology of BACI outcomes. Each panel (a–f) indicates whether each type is associated with a significant (✓) or non-significant (×) result for the main effects (time and location) and the interaction between time and location, or whether either result is possible (~). See Results section entitled *Typology of BACI Outcomes* for further description of each type.

ecosystems (Figure 2). BACI assumes spatial independence among treatments, and often MPA implementation also assumes that outcomes inside will be different from those outside. Perhaps the most common expected outcome of MPA implementation is that abundance, biomass, mean size, or biodiversity will increase inside the MPA, while remaining unchanged or even declining outside, an outcome we refer to as “stockpiling” (Figure 2a). Closed Area I appeared to confer a “stockpiling” response for haddock with significantly greater biomass after the MPA was implemented as compared to the control area at 20–40 km (Table 1, Figure 3d).

It is possible that conditions inside the MPA will remain unchanged, but deteriorate outside (Figure 2b). This might be due to MPA protection “backstopping” against further declines inside the MPA, although it could also be due to displacement of impacts, such as fishing effort, from the area now protected to other locations. The response of yellowtail flounder in Closed

Area I demonstrated a potential backstop effect, as there is an increase in abundance and biomass inside the MPA that counterbalances a decrease in the control area at 20–40 km over the same time period (Table 1, Figure 4b and d). MPA implementation can also conceivably lead to unintended consequences, such as declines in a species within the boundaries of the protected area (Figure 2c). This finding may be the result of misspecification of the control area or may reflect a real process, such as a decline in a prey species due to recovery of a predator. We did not find evidence of declines of yellowtail flounder or haddock associated with MPA implementation. However, unintended consequences are illustrated in the response of haddock abundance in Closed Area II relative to the control area at 4–20 km, where abundance within the control area is of higher magnitude than in the closed area after MPA implementation (Table 1, Figure 3e). Although this response is not statistically significant, the marginal difference may be informative.

Table 1. Results of BACI analysis to evaluate the response of haddock and yellowtail flounder to MPAs, Closed Area I (CAI), and Closed Area II (CAII), compared to control areas at different distances from the MPA boundary (4–20 and 20–40 km).

Species	Closed area	Control area	Prob occurrence						Abundance					
			n (df)	Location:			Pr (>Chisq)	n (df)	Location:			Pr (>Chisq)	Location: period	Pr (>Chisq)
				Location	Period	Location: period			Location	Period	Location: period			
Haddock	CAI	4–20 km	491 (12)	0.00	0.06	0.21	0.65	0.41	0.96	0.56	212 (6)	0.07	0.29	0.05
		20–40 km	609 (12)	0.00	0.06	0.32	0.48	0.41	0.76	0.16	228 (6)	0.00	0.26	0.01
		4–20 km	781 (12)	0.51	0.00	0.02	0.51	0.09	0.54	0.08	442 (6)	0.10	0.01	0.94
Yellowtail flounder	CAII	20–40 km	812 (12)	0.04	0.00	0.01	0.83	0.09	0.95	0.21	442 (6)	0.15	0.02	0.33
		4–20 km	491 (12)	0.00	0.06	0.50	0.77	0.06	0.72	0.02	276 (6)	0.37	0.57	0.44
		20–40 km	609 (12)	0.00	0.06	0.09	0.05	0.06	0.01	0.06	258 (6)	0.48	0.66	0.22
	CAII	4–20 km	781 (12)	0.00	0.00	0.27	0.42	0.00	0.29	0.00	540 (6)	0.01	0.00	0.03
		20–40 km	812 (12)	0.00	0.00	0.00	0.41	0.00	0.74	0.00	506 (6)	0.02	0.00	0.14

Probability of occurrence, relative abundance (number per tow), and biomass (weight per tow) were examined as response metrics.

Outcomes inside and outside of MPAs might also be shaped by processes that operate across MPA boundaries, meaning that the “control” and “impact” treatments are not entirely independent. This can include fishery or other environmental management measures that have effects at large spatial scales (e.g. protection of spawning behaviour or habitat that might have stock-wide benefits to a species) or large-scale environmental variation. The effects of these processes can be similar trends inside and outside of the MPA in either a positive (Figure 2d) or negative (Figure 2e) direction. The parallel, significant positive response of haddock biomass in Closed Area II and both control areas (4–20 and 20–40 km) indicated a large-scale positive influence over time that may be independent of the MPA (Table 1, Figure 3g and h).

If positive large-scale processes are operating alongside a localized stockpiling benefit, then changes inside and outside of the MPA might show similar direction but different rates of change (Figure 2f). However, a similar outcome is possible if spillover of adults or dispersal of larvae from the MPA is occurring, because this would also cause an increase outside of the MPA, but lagged or at a slower rate (Figure 2f). Closed Area II appears to confer a “spillover” response for yellowtail flounder with a significantly greater increase in biomass within the MPA as compared to the control area at 4–20 km (Table 1, Figure 4g). This interpretation of a spillover effect is reinforced by positive responses of yellowtail flounder abundance inside of Closed Area II and the control areas (i.e. significant before: after effect, Table 1, Figure 4e and f), albeit with slight differences in the rate of change over time.

Choice of response variable and control area

The choice of response variable can influence the interpretation of MPA performance. For example, haddock biomass showed a significant, positive response to MPA implementation with an increase inside of Closed Area I compared to the control area (20–40 km; Figure 3d). However, haddock abundance showed a similar, non-significant, increase in both Closed Area I and the control area (20–40 km; Figure 3b). This suggests that the MPA may have had a strong, positive effect on individual growth of haddock, but did not have the same unique impact on survival. Alternatively, survival might be improved inside the MPA, leading to larger average sizes among resident fish, but abundance outside the MPA was supplemented by larval dispersal and/or density-dependent displacement of smaller fish.

In most examples from Georges Bank, the impact of MPAs was robust to alternative control areas defined at different distances from the MPAs. The trends tended to be similar regardless of the spatial scale of the control area, although in some cases the magnitude of response differed. For example, there were similar responses of yellowtail flounder abundance in Closed Area II and control areas, regardless of the distance of the control area from the MPA boundaries (Figure 4e and f). However, the perception of trends in abundance and biomass of yellowtail flounder differed between Closed Area I and control areas at different distance from the MPA boundary. Closed Area I and the control area at 4–20 km showed similar moderate increases in the number and biomass of yellowtail flounder (Figure 4a and c), whereas the control area at 20–40 km exhibited opposite trends as compared to Closed Area I (Figure 4b and d).

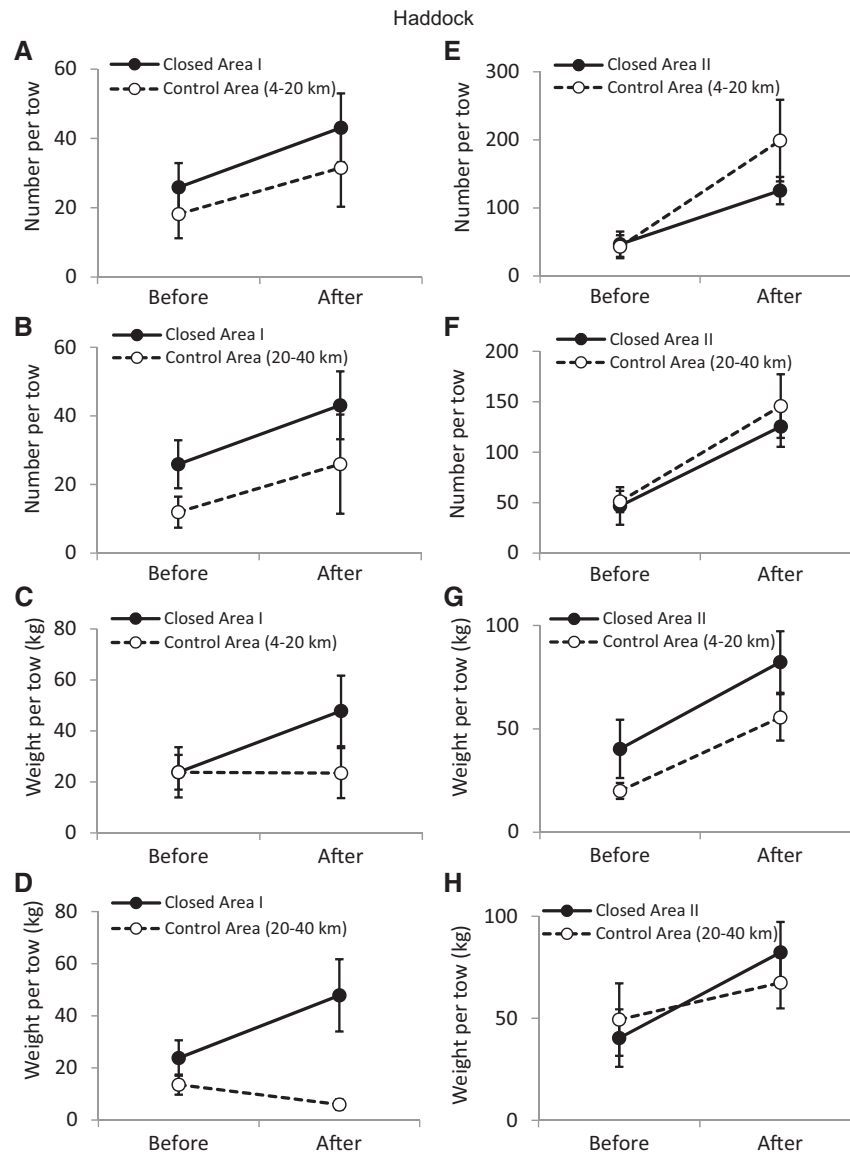


Figure 3. Response of haddock to implementation of MPAs, including relative abundance and weight of haddock inside of Closed Area I (a–d) and Closed Area II (e–h) compared to control areas at different distances from the MPA boundary (4–20 and 20–40 km).

Beyond BACI: temporal and spatial complexity

Temporal patterns of haddock biomass in Closed Area I and yellowtail flounder biomass in Closed Area II illustrate the important signals that can be missed by the binary temporal perspective of BACI (i.e. before–after). In the case of haddock, both the mean and the peaks in biomass were greater inside Closed Area I than the control area at 4–20 km (Figure 5a) after closure. However, the extreme variability in haddock biomass resulted in a non-significant BACI result (Table 1, Figure 3c). In the case of yellowtail flounder, abundance was generally similar inside of Closed Area II and the control area at 4–20 km, however different rates of change support a spillover effect (Figure 4e). Examination of annual patterns in abundance demonstrated that the peaks in biomass of yellowtail flounder in the control area at 4–20 km in 2002 and 2008 lag the peaks within Closed Area II in 2001 and 2007, respectively (Figure 5b). This time lag was reflected in a

significant correlation between Closed Area II and the control area time series (4–20 km) when the control area is lagged by 1 year ($r=0.63$, $p=0.01$), as compared to the relationship when not lagged ($r=0.33$, $p=0.21$).

In addition to the high degree of temporal variability following MPA implementation, there can also be finer-scale spatial variability in the response over time, wherein all habitats within an MPA do not respond uniformly to closure. Following MPA implementation, yellowtail flounder showed recovery, especially within the southeast region of Closed Area II (survey strata 1130 and 1160, Table 2) as well as outside the closed area to the northeast (Figure 6a and b). Yellowtail flounder in Closed Area I showed more modest recovery, but number and biomass indices were both greater after implementation within strata in the middle of the closure (survey strata 1230, Table 2; Figure 6a and b). Haddock demonstrated increases within both Closed Areas I and

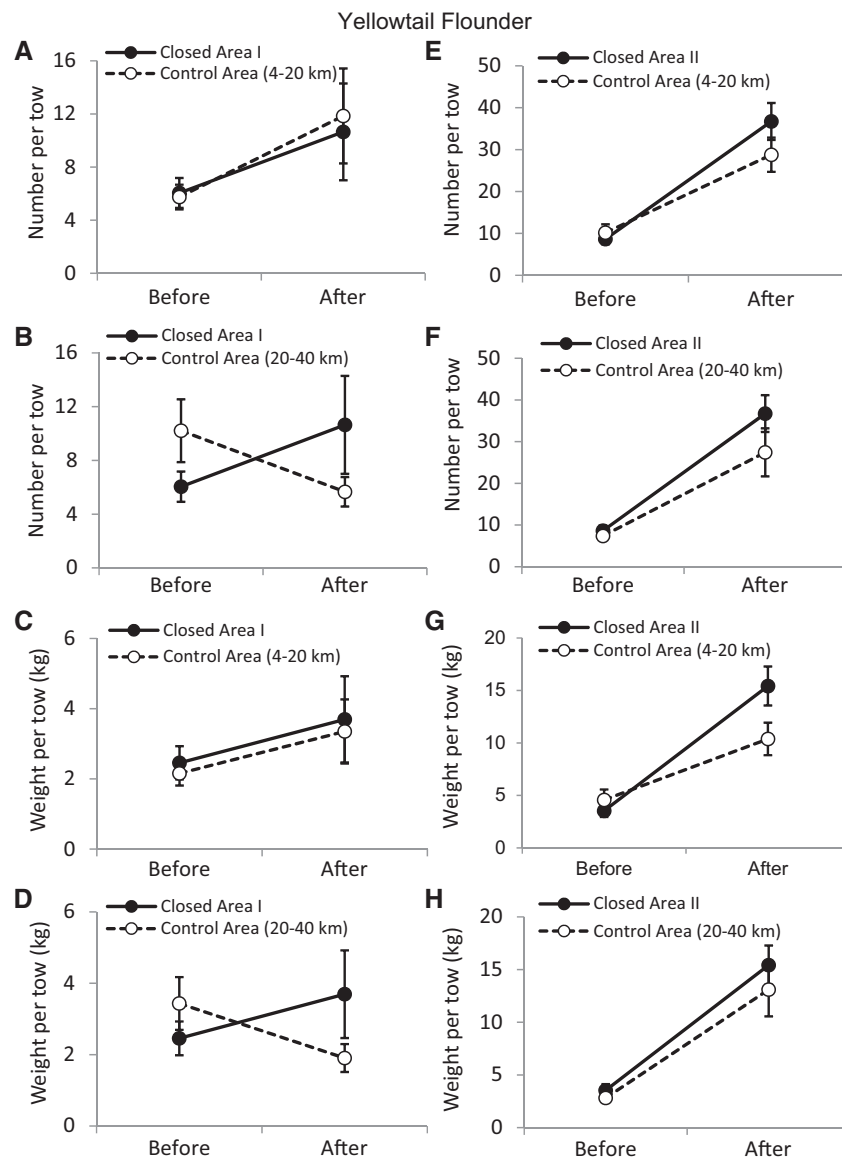


Figure 4. Response of yellowtail flounder to implementation of MPAs, including relative abundance and weight of haddock inside of Closed Area I (a–d) and Closed Area II (e–h) compared to control areas at different distances from the MPA boundary (4–20 and 20–40 km).

II after implementation, with increases in biomass and number in Closed Area I concentrated within the northern region of Closed Area I (strata 1230 and 1240, Table 2, Figure 6c and d), whereas increase in biomass within Closed Area II after implementation were more homogeneously distributed across the MPA (Table 2, Figure 6c and d).

Discussion

Understanding the effectiveness of MPAs is challenging due to the interacting influences of the environment, fisheries, and other resource management measures. However, this is a critical step in quantifying the value of protected areas and their utility as a resource management tool (Pelletier *et al.*, 2005; Pomeroy *et al.*, 2005; Ojeda-Martínez *et al.*, 2011; Fox *et al.*, 2014). Unfortunately, decisions regarding MPAs are often made with

limited understanding of how effective MPAs have been for meeting fishery management objectives, including effects on the productivity of the fish stocks that they were designed to help rebuild.

Our investigation of MPAs on Georges Bank demonstrated the strengths and limitations of applying BACI analyses to the evaluation of MPA effectiveness and the utility of the BACI outcome typology for interpreting MPA impacts. MPAs on Georges Bank were originally designed to protect haddock spawning. Stock assessments indicate substantial increases in haddock biomass across Georges Bank since the mid-1990s (NEFSC, 2017), and our time series and BACI results suggest that this was at least partly due to increases in MPAs. Closed Area I appeared to confer a strong stockpiling effect for haddock, with the increase in Closed Area I contrasted with either unchanged or slight declines in biomass outside of the MPA boundaries depending on which

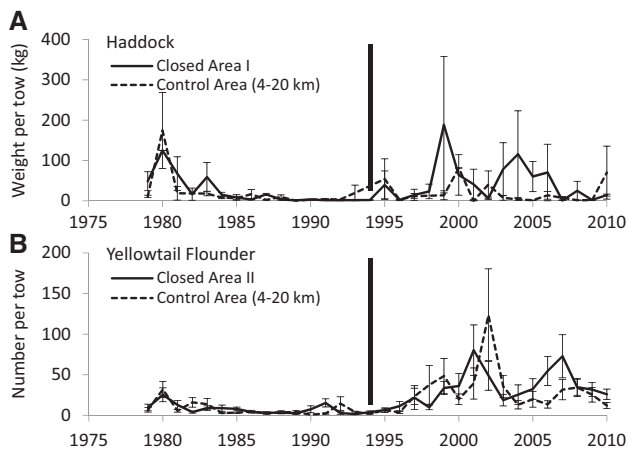


Figure 5. Time series of annual biomass index (weight per tow) of haddock in Closed Area I (a) and abundance index (number per tow) of yellowtail flounder in Closed Area II (b) as compared to control areas (4–20 km). Black vertical line denotes timing of MPA implementation.

control area is considered. However, haddock inside and outside of Closed Area II experienced a common positive response to a large-scale environmental or management change, possibly independent of the MPA or at least sufficiently strong to obscure any MPA effect. Georges Bank haddock have produced several exceptionally strong year classes since the early 2000s (NEFSC, 2017). This increased productivity of haddock has been associated with a link between parental condition and phytoplankton bloom magnitude and phenology such that autumn blooms affect the condition of spawning adults and subsequently recruitment (Friedland *et al.*, 2015). Thus, in the case of haddock, protection of spawning areas in MPAs may have worked in concert with the dominant effect of favourable environmental conditions that supported high recruitment success to rebuild biomass of Georges Bank haddock.

Outcomes for yellowtail flounder showed some similarities with and some important differences from those for haddock. Stock assessments indicate that George Bank yellowtail flounder increased in the early 2000s after MPA implementation, followed by subsequent declines in stock condition since 2010, despite management to decrease exploitation (NEFSC, 2017). BACI and time series results showed that yellowtail flounder increased in number and biomass following MPA implementation through 2010, with either increasing or declining trends outside of the MPA boundaries depending on which control area is considered. Closed Area I appears to serve a backstopping function, protecting a concentration of yellowtail flounder biomass. Localized recovery within Closed Area I is driven largely by changes within a single habitat stratum. Yellowtail flounder recovery within Closed Area II was spread among more habitat strata. Like haddock, the response inside of Closed Area II and the control areas were similar, suggesting common responses to larger scale environmental and/or management change. However, closer examination of the timing of increases suggests that spillover occurred, with peaks of abundance in the control areas lagging Closed Area II. This is consistent with findings by Murawski *et al.* (2005) that suggested a spillover effect of Georges Bank closures in the early 2000s for

yellowtail flounder based on evaluation of fishery-dependent catch per unit effort. Despite indications of recovery in the period after MPA implementation through 2010, subsequent declines have occurred which may be tied with large-scale ecosystem changes in the region. For example, shifts in the distribution of yellowtail flounder have been documented and related to temperature, with the population shifting northward in recent years (Nye *et al.*, 2009). Furthermore, recruitment success has been linked to ocean temperatures with year class strength negatively correlated with bottom temperature (Sissenwine, 1974; Sullivan *et al.*, 2005). These studies indicate a negative effect of climate change on yellowtail flounder in the Northeast US Shelf (Hare *et al.*, 2016) that closed area may not be sufficient buffer.

BACI alone is not suited to detecting complex, yet potentially important, spatial-temporal patterns (Claudet, 2018), and the perception of MPA effectiveness on Georges Bank in some cases was sensitive to the spatial and temporal scales of investigation. For example, examination of data at a finer time-scale, as compared to before-after, enabled evaluation of the precise timing of MPA impacts and identification of potential lags indicative of spillover effects. In addition, finer-scale spatial analysis enabled evaluation of whether MPA effects were achieved evenly inside closures or concentrated within important geographic strata. Without this analysis, MPA benefits might be underestimated because the response can be masked by combining those habitats with other apparently sub-optimal habitats. Such complexity of impacts should be considered in MPA design and evaluation. For instance, because important habitats often differ among species, as is the case for haddock and yellowtail flounder, achieving desired MPA impacts for different species may require protecting a mosaic of habitats (Lindholm *et al.*, 2001). Furthermore, BACI alone may not be the best tool for evaluating benefits produced outside MPAs, as is the objective in some cases. Because BACI looks for contrasts and assumes independence among MPAs or control sites, it is not well suited for detecting effects at broader scales across a fisheries stock or ecosystem. Furthermore, the perception of MPA effectiveness on Georges Bank was sensitive to the choice of response variable and definition of control areas. This supports exploration of multiple response variables and alternative definitions of control areas that explore different hypotheses related to the potential impact of the MPA (i.e. spillover). We evaluated indices of biomass and abundance, but further exploration of other stock attributes (e.g. age and size structure) would be valuable measures of MPA impacts as well.

If MPAs are to be considered as an effective management tactic for conserving marine resources, robust methods for evaluating their effectiveness are essential. BACI is useful in identifying contrasts between MPAs and control areas, changes before and after implementation, and especially the interaction between spatial and temporal effects. However, several aspects of data selection (e.g. response variables) and analytical design (e.g. selection of control areas) need careful consideration, because perceived performance can be influenced by these decisions. In addition, more nuanced interpretation is needed to consider the range of complex effects that are possible (i.e. BACI outcome typology). Ideally, the full analysis would combine BACI analysis with careful inspection of finer resolution temporal trends and spatial patterns to produce a more complete picture of MPA performance.

Table 2. Summary of yellowtail flounder and haddock responses to Closed Areas I and II as compared to control areas at two distances from the MPA boundary (4–20 and 20–40 km).

Species	Location	Survey strata	Before					After				
			N	Abundance	SD	Biomass	SD	N	Abundance	SD	Biomass	SD
Yellowtail flounder	Closed Area I	1130	15	2.28	2.73	0.99	1.16	10	2.60	4.06	0.77	1.23
		1190	15	3.99	8.77	2.42	7.00	13	0.95	1.23	0.19	0.38
		1200	2	0.85	0.00	0.17	0.24	4	0.75	0.96	0.25	0.30
		1230	76	5.21	10.24	1.91	3.38	62	12.38	35.46	4.34	11.93
		1240	23	0.20	0.59	0.12	0.37	15	0.74	1.08	0.29	0.51
	Control area 4–20 km	1130	21	6.97	9.18	2.40	3.24	23	6.56	9.81	1.64	2.03
		1190	40	1.18	2.01	0.59	1.23	18	2.39	6.09	0.51	1.28
		1200	25	1.86	3.50	0.74	1.74	22	1.27	3.07	0.35	0.84
		1230	24	3.63	6.72	1.17	2.15	30	16.94	39.11	5.06	9.95
		1240	22	0.00	0.00	0.00	0.00	31	0.14	0.43	0.05	0.13
	Control area 20–40 km	1130	60	8.32	17.24	2.65	5.36	49	4.43	7.77	1.46	2.68
		1190	36	0.57	1.34	0.31	0.84	35	0.37	0.80	0.11	0.42
		1200	37	0.42	1.11	0.18	0.50	37	0.43	1.26	0.11	0.35
		1230	9	5.04	13.15	0.93	2.18	10	2.33	5.32	0.73	1.36
		1240	55	0.02	0.14	0.01	0.04	46	0.07	0.33	0.02	0.15
	Closed Area II	1130	3	9.07	9.71	3.59	3.88	1	43.88	0.00	20.20	0.00
		1160	116	8.06	13.80	3.13	5.98	135	46.44	65.67	19.55	27.66
		1190	56	1.78	6.95	1.03	4.78	60	3.69	8.28	1.68	3.80
		1200	8	0.00	0.00	0.00	0.00	9	4.38	6.95	0.74	1.19
		1210	2	0.61	0.86	0.13	0.18	8	9.36	11.99	3.16	3.68
	Control area 4–20 km	1130	36	3.91	6.07	1.40	1.97	41	11.62	18.99	3.44	6.05
		1160	72	10.97	23.54	5.04	11.57	81	33.03	45.95	12.19	16.39
		1190	33	0.27	0.59	0.10	0.27	33	1.37	3.61	0.45	1.74
		1200	21	0.23	0.53	0.06	0.14	12	1.62	4.61	0.20	0.45
		1210	26	3.57	7.31	1.52	3.07	28	25.93	68.53	9.88	27.93
	Control area 20–40 km	1130	51	4.83	9.77	1.40	2.64	62	3.72	7.37	2.74	10.22
		1160	52	7.28	9.46	3.07	4.28	55	47.29	78.08	20.86	32.52
		1190	39	0.86	2.33	0.47	1.35	37	0.41	1.45	0.61	3.06
		1200	37	0.61	1.70	0.18	0.76	27	0.62	2.11	0.14	0.62
		1210	28	2.08	4.99	0.71	1.56	26	2.73	4.99	1.24	2.44
Haddock	Closed Area I	1130	15	2.25	4.27	3.01	6.28	10	2.10	4.65	0.87	1.39
		1190	15	5.26	16.06	18.51	58.51	13	0.23	0.83	0.03	0.11
		1200	2	0.41	0.58	0.79	1.12	4	0.00	0.00	0.00	0.00
		1230	76	8.20	25.93	5.50	18.95	62	17.86	51.49	14.38	54.15
		1240	23	37.82	78.04	28.78	64.01	15	102.67	102.25	134.53	164.62
	Control area 4–20 km	1130	21	1.18	1.52	1.76	3.32	23	11.19	51.20	0.75	2.79
		1190	40	10.87	51.07	19.57	74.11	18	0.06	0.24	0.01	0.03
		1200	25	2.33	4.38	6.21	11.65	22	0.14	0.35	0.11	0.49
		1230	24	13.65	24.80	4.70	9.42	30	25.66	74.30	16.03	39.16
		1240	22	1.25	2.94	1.35	2.39	31	13.50	44.76	17.90	71.33
	Control area 20–40 km	1130	60	1.04	2.59	0.77	3.50	49	24.19	113.10	2.32	8.31
		1190	36	0.78	2.71	1.50	6.43	35	0.34	1.53	0.06	0.28
		1200	37	6.67	34.77	6.48	29.99	37	0.09	0.39	0.03	0.19
		1230	9	16.99	46.29	6.26	18.25	10	16.00	34.97	11.57	27.91
		1240	55	3.03	6.76	5.86	13.32	46	1.45	5.38	2.02	6.96
	Closed Area II	1130	3	0.99	1.72	1.36	2.35	1	11.54	0.00	0.90	0.00
		1160	116	18.22	72.41	14.28	45.98	135	94.48	216.50	57.32	159.18
		1190	56	27.88	182.57	23.74	135.56	60	26.23	55.81	29.31	81.47
		1200	8	24.47	57.05	18.21	33.99	9	167.07	277.35	154.38	292.21
		1210	2	1.49	2.11	3.93	5.55	8	229.89	415.26	79.54	76.15
	Control area 4–20 km	1130	36	67.36	304.35	9.11	35.71	41	139.63	637.48	19.01	94.45
		1160	72	25.97	52.69	17.50	36.94	81	156.32	667.92	34.87	77.22
		1190	33	0.74	2.80	1.23	4.02	33	58.32	240.26	27.57	132.89
		1200	21	6.61	25.19	7.23	28.51	12	31.17	67.06	16.08	35.87
		1210	26	12.22	19.72	9.91	17.70	28	135.17	237.30	73.89	139.63
	Control area 20–40 km	1130	51	33.11	102.87	3.92	11.56	62	75.56	361.18	14.74	66.22
		1160	52	20.85	39.90	21.53	48.26	55	127.59	275.07	56.36	129.01
		1190	39	1.06	2.09	2.32	5.25	37	3.32	18.79	3.44	20.69
		1200	37	10.61	46.02	3.41	10.55	27	1.38	5.39	0.78	2.81
		1210	28	91.77	160.67	137.30	349.66	26	199.43	256.67	138.10	171.10

Mean and standard deviation (SD) of abundance and biomass indices within each survey strata are summarized.

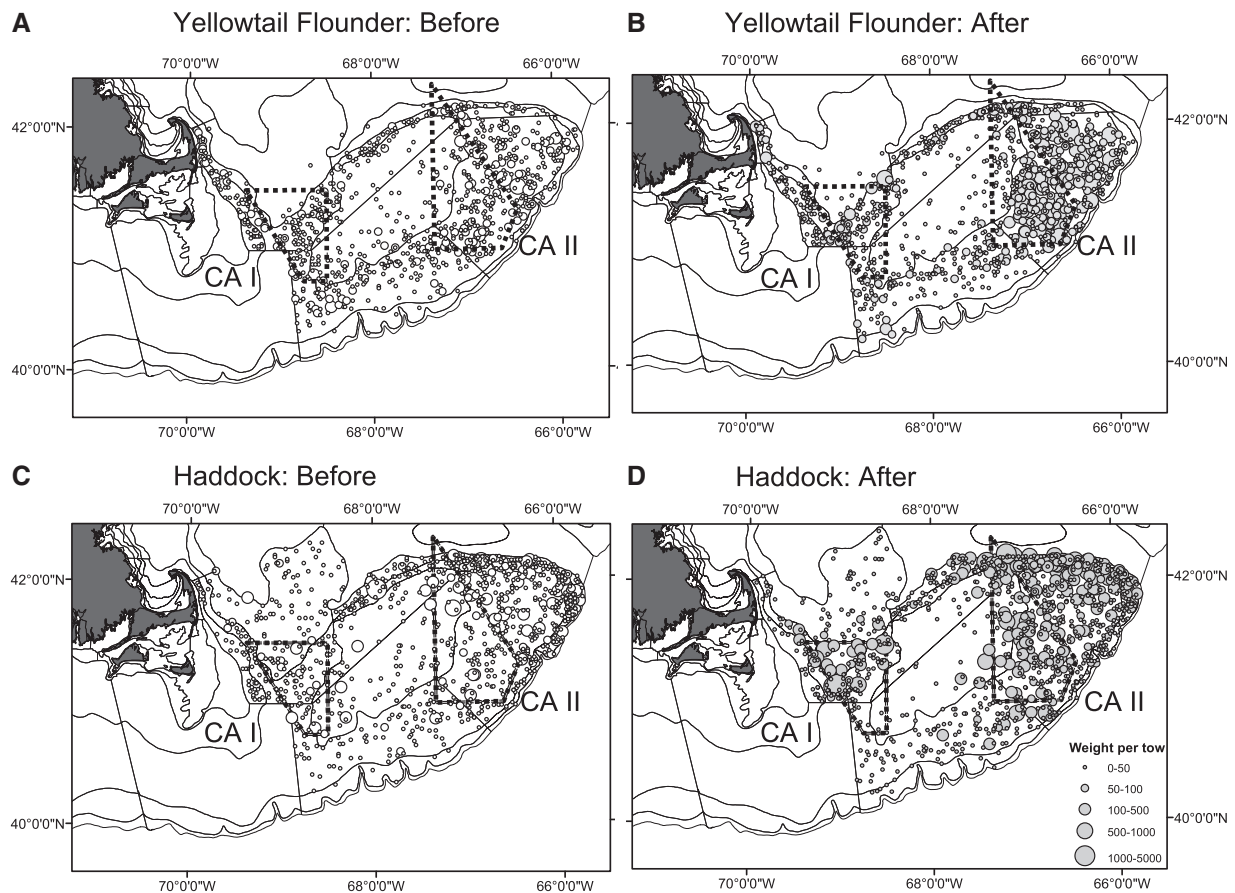


Figure 6. Location of non-zero survey tows and magnitude of biomass per tow for yellowtail flounder (a,b) and haddock (c,d) before (1979–1994) and after (1995–2010) MPA implementation. Note, biomass of survey tows outside of closed area is restricted to the Georges Bank stock area as well as strata inside of closures.

Acknowledgements

We acknowledge the helpful comments and feedback from T. Nies and J. Cournane of the Northeast Fisheries Management Council (NEFMC) on an early version of this manuscript, as well as feedback from the NEFMC Groundfish Plan Development Team and Closed Area Technical Team.

Funding

This work was supported by a grant from the Gordon and Betty Moore Foundation to the Environmental Defense Fund.

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Handling editor: Jonathan Grabowski